STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-SCHEMES-MORPHISMS-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1777-1792 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1794-1856, the complete author proof (61 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_examples-defos.json (pb=2648, stmt 1777-1792, proof 1794-1856).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08U8: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1856–1856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DY7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1425–1440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D0J: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1856–1856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0DY7 let $f : X \to Y$ be a morphism of schemes over k. There is a canonical exact sequence of k-vector spaces*

$$0 \longrightarrow Inf(\mathcal{D}ef_{X\to Y}) \longrightarrow Inf(\mathcal{D}ef_X \times \mathcal{D}ef_Y) \longrightarrow Der_k(\mathcal{O}_Y, f_*\mathcal{O}_X)$$
$$T\mathcal{D}ef_{X\to Y} \longrightarrow T(\mathcal{D}ef_X \times \mathcal{D}ef_Y) \longrightarrow Ext^1_{\mathcal{O}_X}(Lf^* NL_{Y/k}, \mathcal{O}_X)$$

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The obvious map of deformation categories $\mathcal{D}ef_{X\to Y} \to \mathcal{D}ef_X \times \mathcal{D}ef_Y$ gives two of the arrows in the exact sequence of the lemma. Recall that $\mathrm{Inf}(\mathcal{D}ef_{X\to Y})$ is the set of automorphisms of the trivial deformation

$$f' : X' = X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\epsilon]) \xrightarrow{f\times \mathrm{id}} Y' = Y \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\epsilon])$$

of $X \to Y$ to $k[\epsilon]$ equal to the identity modulo ϵ. This is clearly the same thing as pairs $(\alpha, \beta) \in \mathrm{Inf}(\mathcal{D}ef_X \times \mathcal{D}ef_Y)$ of infinitesimal automorphisms of X and Y compatible with f', i.e., such that $f' \circ \alpha = \beta \circ f'$. By Deformation Theory, Lemma 08U8 for an arbitrary pair (α, β) the difference between the morphism $f' : X' \to Y'$ and the morphism $\beta^{-1} \circ f' \circ \alpha : X' \to Y'$ defines an element in

$$\mathrm{Der}_k(\mathcal{O}_Y, f_*\mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_Y}(\Omega_{Y/k}, f_*\mathcal{O}_X)$$

Equality by More on Morphisms, Lemma 0D0J. This defines the last top horizontal arrow and shows exactness in the first two places. For the map

$$\mathrm{Der}_k(\mathcal{O}_Y, f_*\mathcal{O}_X) \to T\mathcal{D}ef_{X\to Y}$$

we interpret elements of the source as morphisms $f_\epsilon : X' \to Y'$ over $\mathrm{Spec}(k[\epsilon])$ equal to f modulo ϵ using Deformation Theory, Lemma 08U8. We send f_ϵ to the isomorphism class of $(f_\epsilon : X' \to Y')$ in $T\mathcal{D}ef_{X\to Y}$. Note that $(f_\epsilon : X' \to Y')$ is isomorphic to the trivial deformation $(f' : X' \to Y')$ exactly when $f_\epsilon = \beta^{-1} \circ f \circ \alpha$ for some pair (α, β) which implies exactness in the third spot. Clearly, if some first order deformation $(f_\epsilon : X_\epsilon \to Y_\epsilon)$ maps to zero in $T(\mathcal{D}ef_X \times \mathcal{D}ef_Y)$, then we can choose isomorphisms $X' \to X_\epsilon$ and $Y' \to Y_\epsilon$ and we conclude we are in the image of the south-west arrow. Therefore we have exactness at the fourth spot. Finally, given two first order deformations X_ϵ, Y_ϵ of X, Y there is an obstruction in

$$ob(X_\epsilon, Y_\epsilon) \in \mathrm{Ext}^1_{\mathcal{O}_X}(Lf^* NL_{Y/k}, \mathcal{O}_X)$$

which vanishes if and only if $f : X \to Y$ lifts to $X_\epsilon \to Y_\epsilon$, see Deformation Theory, Lemma 08U8. This finishes the proof. □